

RIGIDITY AND UNIVERSAL CONTRACTIONS OF HARD-SPHERE KINETIC COTANGENTS

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ABSTRACT. A phase is introduced as a genuine positive latent variable with conditional microscopic path laws. The joint labelled experiment has a good large-deviation principle and contracts to the minimum of the phase costs and component rates. Restricted pressures use one consistent sign and the exact finite-volume value is a log-sum-exp. Dynamic consistency is restored by carrying the adapted phase posterior in the state; the marginal log-sum alone is not called a semigroup. Quantitative complex dominance gives phase-local response charts, and second-order component weights determine labelled Gaussian mixtures at coexistence.

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1. ROUND-SEVENTEEN POSITIVE CLOSURE: LATENT PHASES, ADAPTED POSTERIOR DYNAMICS, AND NONLINEAR AGGREGATION

A phase is a genuine positive latent variable in the microscopic probability space, not a smooth partition weight renamed as a discrete event. Dynamic consistency is obtained by carrying its posterior distribution as part of the

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state. The marginal log-sum value is then the projection of an augmented semigroup; it is not asserted to be a semigroup on the physical state alone.

1.1. Positive latent-phase experiments. Let $J \in \{0, \dots, m\}$ be sampled at time zero with positive weights $w_{\varepsilon,j}$. Conditional on $J = j$, the microscopic path has law $P_{\varepsilon,j}$ on a common measurable path space. In the Sinai platform these conditional laws are obtained from positive recurrent components of the inducing graph plus a boundary component. In the hard-sphere platform they are the positive source/preparation phases constructed by B1–B4. The physical law is exactly

$$P_\varepsilon = \sum_{j=0}^m w_{\varepsilon,j} P_{\varepsilon,j}.$$

No spectral projection is used as a probability.

Theorem 1.1 (Joint labelled large deviations). *Assume each positive conditional experiment satisfies a good path LDP at speed a_ε with rate I_j , and*

$$-a_\varepsilon^{-1} \log w_{\varepsilon,j} \rightarrow \alpha_j.$$

Then (J, X_ε) satisfies a good LDP on the discrete-path product with rate

$$\mathcal{I}(j, x) = \alpha_j + I_j(x) - \min_k \alpha_k.$$

The physical marginal has rate

$$I(x) = \min_j \{\alpha_j + I_j(x)\} - \min_k \alpha_k.$$

The component rates are those of genuine conditional laws, not envelopes defined after the fact.

Proof. For a closed set F in the finite product,

$$P_\varepsilon((J, X) \in F) = \sum_j w_{\varepsilon,j} P_{\varepsilon,j}(F_j).$$

The largest exponential term gives the upper bound; the corresponding open slice gives the lower bound. Exponential tightness is inherited from the finite family. Contraction over j gives the marginal rate. The formula uses actual positive conditional probabilities throughout. $\square$

1.2. Restricted pressure with one sign convention. Define the normalized component pressure

$$\tilde{Q}_{\varepsilon,j}(F) = \frac{1}{a_\varepsilon} \log E_{\varepsilon,j} e^{a_\varepsilon F}$$

and the unnormalized contribution

$$Q_{\varepsilon,j}^{\text{un}}(F) = \frac{1}{a_\varepsilon} \log E_\varepsilon [e^{a_\varepsilon F}; J = j].$$

Lemma 1.2 (Exact phase-cost sign). *With $\alpha_{\varepsilon,j} = -a_{\varepsilon}^{-1} \log w_{\varepsilon,j}$,*

$$Q_{\varepsilon,j}^{\text{un}}(F) = -\alpha_{\varepsilon,j} + \tilde{Q}_{\varepsilon,j}(F).$$

Consequently

$$Q_{\varepsilon}(F) = \frac{1}{a_{\varepsilon}} \log \sum_j \exp \left\{ a_{\varepsilon} [-\alpha_{\varepsilon,j} + \tilde{Q}_{\varepsilon,j}(F)] \right\}$$

and the phase cost is counted exactly once.

Proof. The first identity is $E[e^{aF}; J = j] = w_j E_j[e^{aF}]$ followed by a logarithm. Summing the exact positive contributions gives the second formula. $\square$

1.3. Adapted posterior phase state. Let $\mathcal{F}_t$ be the observed filtration. The phase posterior is

$$\rho_t(j) = P(J = j \mid \mathcal{F}_t).$$

If the component transition kernels have likelihood increments $L_{s,t}^j$, Bayes' rule gives

$$\rho_t(j) = \frac{\rho_s(j) L_{s,t}^j}{\sum_k \rho_s(k) L_{s,t}^k}.$$

This process is adapted and automatically consistent under horizon restriction.

Theorem 1.3 (Augmented phase-posterior semigroup). *Let S_t^j be the component risk-sensitive transition semigroup, including its physical/filter state. On the augmented state (x, ρ) define $\mathbf{S}_t$ by evolving each component and updating ρ with its likelihood increment. Then $(\mathbf{S}_t)$ is a nonlinear semigroup. For terminal payoff h its marginal value is the exact nonlinear aggregation*

$$\mathcal{V}_t(x, \rho; h) = \frac{1}{\theta} \log \sum_j \rho(j) \exp \{ \theta S_t^j h(x) \}$$

when the component likelihood and risk coefficient are typed in the same canonical experiment. The map $x \mapsto \mathcal{V}_t$ alone need not form a semigroup; the augmented pair does.

Proof. Condition on $(J, \mathcal{F}_s)$. Each component obeys its own tower law, and Bayes' update of ρ at time s followed by that at time $s + t$ is identical to the one-step update because likelihood ratios multiply. This proves the augmented semigroup law. Summing the positive conditional exponential values gives the displayed log-sum formula. Retaining the posterior is exactly the state variable missing from a naive marginal log-sum operator. $\square$

1.4. Phase-local analytic charts. For a component j , let K_j be a positive transfer/coefficient operator on the recurrent interior of that component. Assume its simple leading eigenvalue is separated by a gap and that the leakage to other components is exponentially small on a complex source neighborhood.

Theorem 1.4 (Quantitative zero-free phase charts). *There are component neighborhoods U_j on which the restricted partition function is zero free and $\tilde{Q}_j$ is analytic. More precisely, if*

$$Z_{\varepsilon,j}(F) = c_{\varepsilon,j}(F)e^{a_{\varepsilon}q_j(F)} + R_{\varepsilon,j}(F),$$

with $|c| \geq c_0$ and $|R| \leq \frac{1}{2}c_0e^{a_{\varepsilon}\Re q_j}$ on U_j , then $Z_{\varepsilon,j}$ has no zero there and all pressure derivatives converge locally uniformly. These estimates are supplied by the A2/A3 spectral component or B1/B2 positive coefficient component; positivity on the real axis alone is not used.

Proof. On the boundary of each compact subchart, the remainder is strictly smaller than the leading analytic term. Rouché's theorem gives zero freeness and the same number of zeros as the nonvanishing leading term, namely none. A consistent analytic logarithm exists, and Cauchy's formula gives convergence of all derivatives. The spectral/coefficient gap is a quantitative complex estimate, not a consequence of real positivity. $\square$

1.5. Second-order labelled limits at coexistence. Assume

$$w_{\varepsilon,j} = e^{-a_{\varepsilon}\alpha_j} a_{\varepsilon}^{-\gamma_j/2} (c_j + o(1))$$

and, under component j ,

$$\sqrt{a_{\varepsilon}}(TX_{\varepsilon} - m_j) \Rightarrow N(0, \Sigma_j).$$

Theorem 1.5 (Labelled Gaussian mixtures and tie weights). *At a unique minimizing phase, the usual component Gaussian and Cameron–Martin tangent holds. At a tie set J_* , the joint labelled field converges to the mixture*

$$\sum_{j \in J_*} \omega_j \delta_j \otimes N(m_j, \Sigma_j), \quad \omega_j = \frac{c_j \mathbf{1}_{\{\gamma_j = \gamma_*\}}}{\sum_{k \in J_*} c_k \mathbf{1}_{\{\gamma_k = \gamma_*\}}}.$$

Conditioning sequences with different subexponential asymptotics produce the corresponding explicitly tracked posterior weights; no single covariance or Cameron–Martin space is assigned to an unlabeled tie mixture.

Proof. Apply the component central limit theorem jointly with the discrete label. Laplace's method in the exact finite sum selects the phases with minimum exponential cost and then the minimum polynomial exponent; the coefficients c_j normalize the remaining weights. Keeping the label gives joint weak convergence. Marginalization yields the Gaussian mixture, while conditional Cameron–Martin shifts act componentwise. $\square$

1.6. Typed contractions and calibration.

Theorem 1.6 (Phase-aware contraction theorem). *For every typed observable map T , the labelled rate contracts to*

$$I_T(y) = \min_j \inf_{Tx=y} \{\alpha_j + I_j(x)\} - \min_k \alpha_k.$$

Conditioning, component pressure differentiation, Gaussian tangent formation, and posterior aggregation commute on the augmented labelled state whenever the upstream component interfaces hold. A mechanical preparation multiplier is identified with a risk-sensitive coefficient only under an explicit canonical-cocycle compatibility axiom; phase decomposition alone does not make that identification.

Proof. Apply the contraction theorem to the joint rate of Theorem 1.1 and then minimize over the finite label. Componentwise analytic differentiation and Gaussian convergence commute with T by the upstream typed contraction theorem of C2. Bayes aggregation is an exact finite conditional-expectation identity, so it commutes on the augmented state. Finally, equality of the mechanical and valuation exponents is an extra equality of Radon–Nikodym cocycles; without it the two scalar parameters are independent. $\square$

Theorem 1.7 (Round-seventeen D1 closure). *Positive latent phases, their conditional path laws, adapted posterior updates, component LDPs, zero-free response charts, nonlinear augmented semigroups, and labelled Gaussian mixtures form one causal and horizon-consistent phase theory. Phase costs have one sign and are counted once; the standalone theorem does not rely on signed spectral labels or a non-semigroup marginal log-sum operator.*

Proof. Combine Theorems 1.1, 1.3, 1.4, 1.5, and 1.6, together with Lemma 1.2. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Seventeen dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

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